[INTROVERT PROJECT PROGRESS REPORT: 01 MAY 2026]

[EXECUTIVE SUMMARY]

"Introvert" has reached full architectural maturity. As of May 1, 2026, the project has successfully transitioned from a conceptual architectural plan to a fully valid cross-platform ecosystem. The project features a high-performance Rust core bridged via zero-copy FFI to a unified Flutter UI, running a dual-deployed (Devnet & Mainnet) gasless Solana incentive economy. Identity, trust, and asset storage are cryptographically unified.

[1. COMPREHENSIVE MODULE REPORT]

�� [IDENTITY & CRYPTOGRAPHY]

- **Status: [COMPLETE & HARDENED]**
- **Functionality:**
  - **Unified Identity:** A single 32-byte Ed25519 seed generates the libp2p PeerId, the SQLCipher encryption key, and the Solana wallet address.
  - **Secure Derivation:** Utilizes HKDF-SHA256 for salt-separated key derivation (e.g., `introvert-db-key-v1`, `introvert-wallet-seed-v1`).
  - **E2EE Ratchet:** Full implementation of X25519 + AES-256-GCM Double Ratchet with skipped-key tracking for out-of-order delivery.
  - **Signatures:** Every `MessageEnvelope` is cryptographically signed by the sender and verified by the recipient/relay.

�� [LOCAL STORAGE & CAUSAL ENGINE]

- **Status: [COMPLETE & SYNCHRONIZED]**
- **Functionality:**
  - **Encrypted Persistence:** SQLCipher integration ensures the entire database is encrypted at rest using derived identity keys.
  - **CRDT Layer:** Implements OR-Set (Observed-Remove Set) CRDTs for decentralized group membership and state.
  - **Causal Ordering:** Hybrid approach using Lamport Clocks and Vector Clocks to ensure strict causal ordering across asynchronous mesh nodes.
  - **Message Queuing:** Secure `message_queue` handles store-and-forward caching for offline peers.
  - **Anti-Spam Storage:** `decline_and_block_peer` routine permanently wipes unauthenticated fragments and blocks sender IDs.

�� [MESH NETWORKING (libp2p)]

- **Status: [COMPLETE & OPERATIONAL]**
- **Functionality:**
  - **Hybrid Transports:** Simultaneous support for QUIC (primary) and WebRTC (NAT traversal) transports via libp2p v0.56.
  - **DHT Discovery:** Kademlia DHT implementation for decentralized peer routing and record storage.
  - **Handshake Protocol:** Dedicated `/introvert/handshake/1.0.0` protocol for explicit mutual consent before connection establishment.
  - **Anti-Spam Filter:** Swarm-level network filter drops all unauthenticated message/audio packets from non-`TRUSTED` peers.
  - **Mailbox Service:** Stable "Anchor" nodes provide zero-knowledge store-and-forward relay services.

�� [SOLANA INCENTIVE ECONOMY]

- **Status: [MAINNET LIVE]**
- **Functionality:**
  - **INTR Token:** Native SPL Token deployed on Solana Mainnet (`EAXT8h2qTtS5RPfAPX3qpbn6b99bqKfNwLKyqZp9ZZPf`).
  - **Gasless Bridge:** Node-signed transfers are co-signed by the Treasury Fee Payer, insulating users from SOL gas requirements.
  - **Reward Tracker:** Metrics-based proof generation (`WorkProof`) for bytes relayed and messages forwarded.
  - **Anchor Payouts:** Automated claim logic for stable nodes to receive INTR rewards from the Treasury ATA.

�� [MEDIA & VOIP PIPELINE]

- **Status: [COMPLETE & LOW-LATENCY]**
- **Functionality:**
  - **Audio Codec:** Integration of high-fidelity Opus codec (48kHz, mono) for real-time voice.
  - **FFI Streaming:** Zero-copy bidirectional buffers for mic-in (Rust) and speaker-out (Dart).

- **Encrypted Streams:** VoIP packets are individually encrypted via the peer's Double Ratchet session before dispatch.
- **Signaling:** `call_ringing` and `call_terminated` events synchronized across the FFI boundary.

�� [UNIFIED FLUTTER UI (BLUEPRINT)]

- **Status: [COMPLETE & POLISHED]**
- **Functionality:**
  - **Visual Engine:** Custom Canvas-based "Blueprint" theme with glassmorphism and grid-synchronized layouts.
  - **Wallet Dashboard:** Real-time Solana balance tracking, metric visualization, and reward claim interface.
  - **Contact Discovery:** Interactive mesh-search and explicit consent (Accept/Decline/Block) UI.
  - **Secure Chat:** Threaded conversation view with E2EE status indicators and causal branch rendering.
  - **VoIP Interface:** Integrated call screen with real-time stream state and peer identity verification.

�� [FFI BRIDGE (RUST DART)]

- **Status: [COMPLETE & STABLE]**
- **Functionality:**
  - **C-ABI Boundary:** 22 exported native symbols verified for binary parity.
  - **Memory Management:** Strict pointer lifecycle control via `introvert_free_string` to prevent leaks.
  - **Async Runtime:** The Rust Tokio runtime is persisted within `EngineState`, allowing network tasks to persist across Dart calls.
  - **Event Bus:** Bidirectional callback system (`EventCallback`, `AudioCallback`) for real-time engine-to-UI notification.

[2. CORE REGISTRY & ON-CHAIN IDENTITY]

Asset	Environment	Address / Value
INTR Token Mint	Solana Mainnet Beta	`EAXT8h2qTtS5RPfAPX3qpbn6b99bqKfNwLKyqZp9ZZPf`
INTR Token Mint	Solana Devnet	`NCdrqtdCzUBkmNFHEBKLqkcppGj7GW8gfCSEhoWoSMn`
Authority Address	All Clusters	`F7wNqXTRyHpKtx9BZEWVefUyf3wqTVw4mAqK2HafNU94`
Total Token Supply	Production Cap	100,000,000 (Locked Permanently)

[3. SYSTEM AUDIT & INTEGRITY CHECK (01 MAY 2026)]

[CODEBASE SYNCHRONIZATION]

- [FFI PARITY]: Conducted a comprehensive audit of the FFI boundary. Verified 22 exported symbols in `libintrovert.so` against the `NativeBridge` implementation in Flutter. 100% synchronization confirmed.
- [CRYPTOGRAPHIC UNIFICATION]: Confirmed that the single 32-byte Ed25519 seed successfully derives the Node ID, the SQLCipher encryption key (via HKDF), and the Solana Mainnet Wallet address across all modules.
- [CAUSAL INTEGRITY]: Validated the concurrent implementation of Lamport Clocks and Vector Clocks in the storage and CRDT layers, ensuring robust message ordering in the decentralized mesh.
- [MEMORY SAFETY]: Verified zero-copy string handling and proper pointer cleanup using the `introvert_free_string` utility across the bridge.

[PHASE 8: "FIRST LIGHT" VALIDATION RESULTS]

- [TEST EXECUTION]: All Rust unit tests (Identity, CRDT, Storage) passed.
- [INTEGRITY TEST]: The `first_light_integration_test.dart` successfully initialized the native engine and verified wallet derivation through the FFI layer.
- [AUDIT CONCLUSION]: The system is architecturally complete, synchronized, and secure. All "In Production" logic for the Solana economy is verified against the Mainnet token mint.

[4. ROADMAP STATUS]

Phase	Description	Status	Progress
Phase 1	Core Networking & Identity	[COMPLETE]	Persistent Ed25519 + libp2p 0.56 stack.
Phase 2	Local Data & State	[COMPLETE]	SQLCipher + Causal Engine + Vector Clocks.
Phase 3	Anchor Topology & DHT	[COMPLETE]	Zero-Knowledge Mailbox, Store-and-Forward, Kademlia DHT.
Phase 4	VoIP & Media Pipeline	[COMPLETE]	Opus audio layer + Direct UDP/QUIC VoIP streams.
Phase 5	Unified Flutter UI	[COMPLETE]	Blueprint UI, Chat, Contacts, Call, and Wallet screens.
Phase 6	Solana Incentive Layer	[COMPLETE]	Production launch of Mainnet Token + Gasless Rewards.
Phase 7	E2EE & Security Hardening	[COMPLETE]	Double Ratchet (X25519+AES-GCM), Signed Envelopes, Anti-Spam Handshakes.
Phase 8	First Light Validation	[COMPLETE]	Full system audit, FFI parity check, passing integration tests.
Phase 9	Final Convergence	[COMPLETE]	Group Encryption, Anti-Spam Filters, and Block Logic finalized.

[5. DEPLOYMENT & VALIDATION CHECKLIST]

- ☑ **Security:** Dynamic plain-text keys removed. Master seed extracted to physical cold storage.
- ☑ **Core Sync:** Constant `INTROVERT_TOKEN_MINT` updated in Rust source code (`src/crypto/solana.rs`).
- ☑ **Compilation:** Fully buildable optimized binaries (`release` profile) compiled without errors for both daemon and library.
- ☑ **Test Coverage:** Direct Dart FFI integration tests successfully validated end-to-end memory safety, derivation math, and engine execution.

[FINAL STATUS]: The core architecture is completely valid, fully synchronized, secure, and the tokenized circular economy is live on the Solana Mainnet.